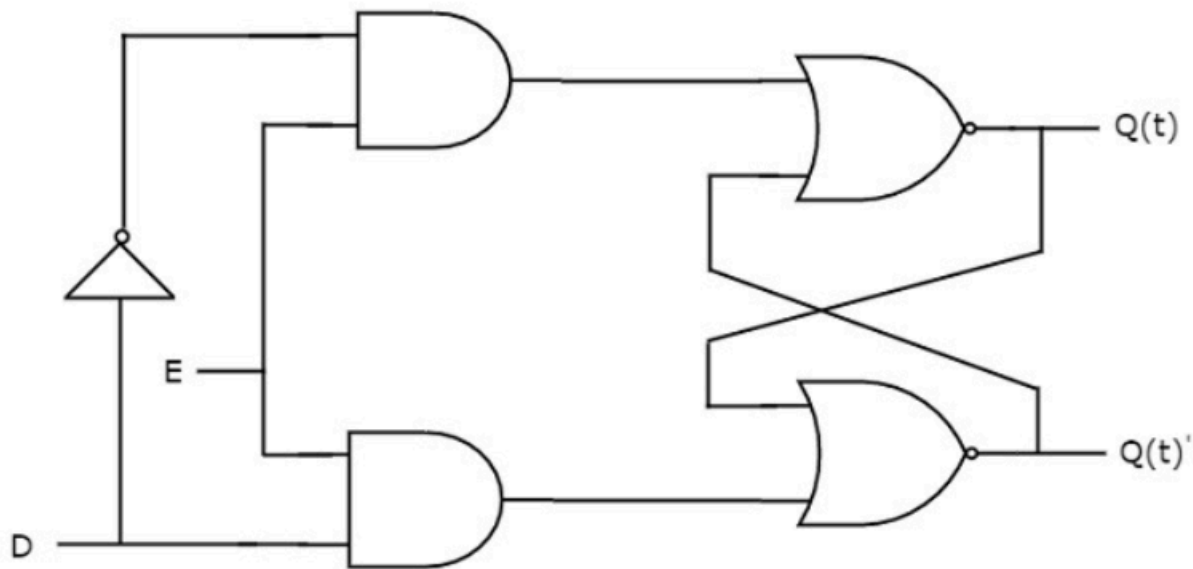


Circuit Simulation Lab:11

Design of Latches

The purpose of this experiment is to introduce the design of simple sequential circuits, in this case Latches. Latch is an electronic device that can be used to store one bit of information. The D latch is used to capture, or 'latch' the logic level which is present on the Data line when the clock input is high. If the data on the D line changes state while the clock pulse is high, then the output, Q, follows the input, D. When the CLK input falls to logic 0, the last state of the D input is trapped and held in the latch.

Code:



Dataflow modeling

Code:

```
module dl_data(  
input e , d ,  
output q , qbar );  
  
assign q = (e & d ) | ( ~e & q );  
assign qbar = ~((e & d ) | ( ~e & q ));  
  
endmodule
```

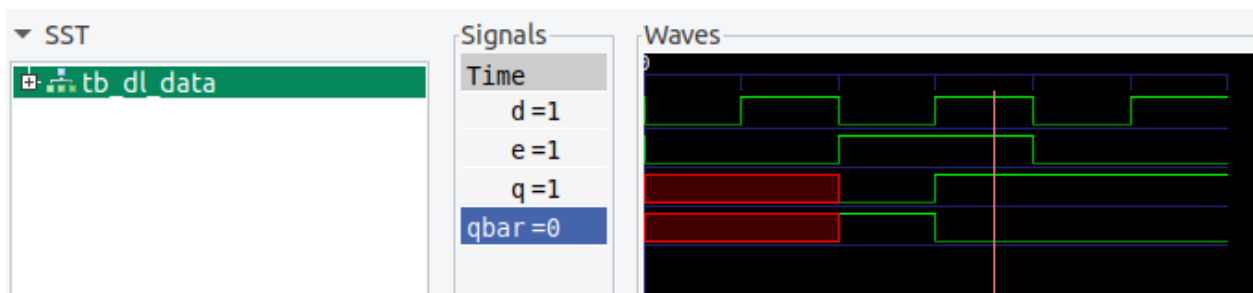
TESTBENCH

```
module tb_dl_data ;
reg e , d ;
wire q , qbar;

dl_data d1 (
.e(e),.d(d),.q(q),.qbar(qbar));

initial begin
$monitor ("e = %b , d =%b , q = %d , qbar = %b,", e, d, q , qbar );
end

initial begin
$dumppfile ("df.vcd");
$dumppvars (0,tb_dl_data);
e = 0 ; d = 0 ; #10;
e = 0 ; d = 1 ; #10;
e = 1 ; d = 0 ; #10;
e = 1 ; d = 1 ; #10;
e = 0 ; d = 0 ; #10;
e = 0 ; d = 1 ; #10;
end
endmodule
```



```
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASS1
VCD info: dumpfile df.vcd opened for output.
e = 0 , d =0 , q = x , qbar = x,
e = 0 , d =1 , q = x , qbar = x,
e = 1 , d =0 , q = 0 , qbar = 1,
e = 1 , d =1 , q = 1 , qbar = 0,
e = 0 , d =0 , q = 1 , qbar = 0,
e = 0 , d =1 , q = 1 , qbar = 0,
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASS1
```

```

BEHAVIOURAL;
module dl_beh(
input e,
input d,
output reg q,
output qbar
);

```

```

always @(*)
begin
if(e)
    q = d;
end

```

```

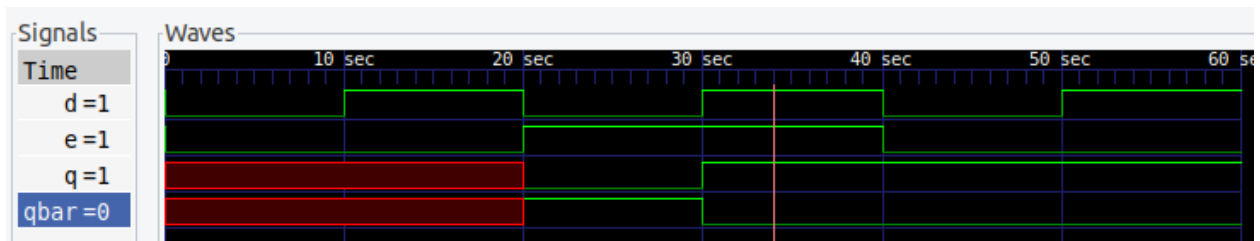
assign qbar = ~q;

```

```

Endmodule

```



```

STRUCTURAL:

```

```

module dl_structural(
input e,
input d,
output q,
output qbar
);

```

```

wire nd;
wire s;
wire r;

```

```

not g1(nd,d);

```

```

and g2(s,d,e);
and g3(r,nd,e);

```

```

nor g4(q,r,qbar);
nor g5(qbar,s,q);

```

endmodule

